Week 3 Tutorial – Problem Solving Challenge

Step 1: Understand and Define the Problem (Analyse)

The problem here requires designing a combinational logic circuit that triggers an alarm in a car whenever the seat in a car is occupied and are not fastened. The circuit must have two active-HIGH signals DRIV and PASS to identify the occupant representing the driver and passenger respectively. If the car IGN is ON and none of the seats are unfastened, the alarm must be activated (LOW).

Step 2: Organise and Describe the Data

Symbol	Meaning	Type	Active Logic
DRIVE	Driver in the seat	Input	HIGH or 1
PASS	Passenger in the seat	Input	HIGH or 1
$\overline{BELTD}$	Driver's seatbelt unfastened	Input	LOW or 1
$\overline{BELTP}$	Passenger's seatbelt unfastened	Input	LOW or 0
IGN	Ignition switch is ON	Input	HIGH or 1
ALARM	Alarm sound triggered	Output	LOW or 0

Step 3: Plan the Solution (Design the Algorithm)

3.1 Algorithm (in plain English)

1. Check if the IGN is HIGH or LOW. The ignition switch must be HIGH.
2. If driver is in the seat and seatbelt is fastened, then the DRIV = 1 and BELTD = 0.
3. If passenger is in the seat and the seatbelt is not fastened, then the PASS = 1 and BELTP = 0.
4. If either condition above is TRUE and IGN = 1, the alarm gets triggered, LOW.
5. Otherwise, the alarm remains HIGH.

3.2 Truth Table

INPUT					OUTPUT
IGN	DRIV	BELTD	PASS	BELTP	ALARM
1	0	0	0	0	1
1	0	0	0	1	1
1	0	0	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	0	1	0
1	1	0	1	0	0
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	0	1	1
1	1	1	1	0	0
1	1	1	1	1	1

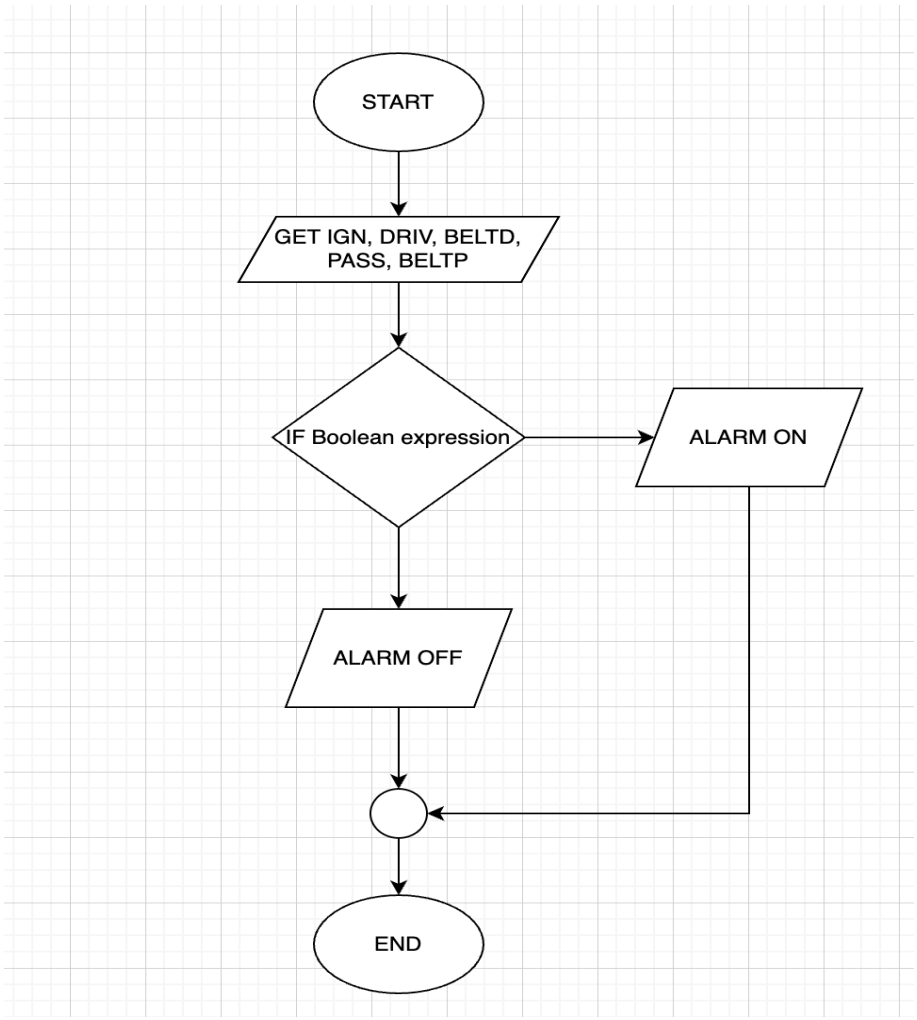
ALARM triggered = 0, ALARM not triggered = 1.

3.3 Boolean Expressions (SOP)

1. DRIV.$\overline{BELTD}$.$\overline{PASS}$.$\overline{BELTP}$.IGN
2. DRIV. $\overline{BELTD}$.$\overline{PASS}$.BELTP.IGN
3. DRIV. $\overline{BELTD}$.PASS.$\overline{BELTP}$.IGN
4. DRIV. $\overline{BELTD}$.PASS.BELTP.IGN
5. DRIV,BELTD,PASS.$\overline{BELTP}$.IGN

ALARM = DRIV.$\overline{BELTD}$.$\overline{PASS}$.$\overline{BELTP}$.IGN + DRIV. $\overline{BELTD}$.$\overline{PASS}$.BELTP.IGN + DRIV. $\overline{BELTD}$.PASS.$\overline{BELTP}$.IGN + DRIV. $\overline{BELTD}$.PASS.BELTP.IGN + DRIV.BELTD.PASS.$\overline{BELTP}$.IGN

3.4 Flowchart



3.5 Pseudocode

START

GET IGN, DRIV, BELTD, PASS, BELTP

IF DRIV.$\overline{BELTD}$.$\overline{PASS}$.$\overline{BELTP}$.IGN + DRIV. $\overline{BELTD}$.$\overline{PASS}$.BELTP.IGN + DRIV. $\overline{BELTD}$.PASS.$\overline{BELTP}$.IGN + DRIV. $\overline{BELTD}$.PASS.BELTP.IGN + DRIV.BELTD.PASS.$\overline{BELTP}$.IGN

THEN ALARM ON

ELSE ALARM OFF

END

Step 4: Test and Refine the Solution (Debug and Verify)

1. All the seats are unoccupied.

IGN	DRIV	BELTD	PASS	BELTP	EXPECTED ALARM
1	0	0	0	0	1

Here, no driver or the passenger is present in the car, hence, the alarm won't be triggered when the ignition is ON.

2. The driver is present in the seat and the seatbelt is unfastened.

IGN	DRIV	BELTD	PASS	BELTP	EXPECTED ALARM
1	1	0	0	0	0

Here the driver is present in the car, and the seatbelt is unfastened, hence, the alarm gets triggered when the ignition is ON.

3. The passenger is present in the seat and the seatbelt is fastened.

IGN	DRIV	BELTD	PASS	BELTP	EXPECTED ALARM
1	0	0	1	1	1

Here, the passenger is only present in the car, and the seatbelt is fastened, so, the alarm won't be triggered when the ignition is ON.

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